LESSON PLAN

AUXILIARY EQUIPMENT (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principles & operations of Super chargers, description and operations
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principles & operations of Super chargers, description and operations
5. Specific Objectives	22.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of engine supercharger diagrams; Safety: No live equipment; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a supercharger in a piston engine?
(b) How does a supercharger differ from a turbocharger regarding power source?
(c) Why is supercharging necessary for high-altitude flight performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Working principle of piston engines • Induction system components • Atmospheric pressure effects • Air-fuel mixture requirements	L & Ds	PPT, IB	15 min
(b)	• Description of supercharger types • Centrifugal supercharger operation • Positive displacement supercharger operation • Drive mechanisms and gear ratios	L & Ds	PPT, PDF	20 min
(c)	• Supercharger operational cycles • Boost pressure regulation • Heat management systems • Maintenance and inspection points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a centrifugal supercharger increases manifold pressure.
(b) Describe the relationship between engine RPM and supercharger output.
(c) Identify three critical maintenance checks for a supercharger system.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational principles of LP 02 (Piston Engine Principles) with the supercharging mechanisms discussed in this lesson.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Terminology used i.e. BDC TDC Stoke length, clearance volume, swept volume, valve lead, valve lag, ineffective crank angle	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min